

## Formation and Combustion

**Subject & topics:** Chemistry | Physical | Energetics      **Stage & difficulty:** A Level P1

For which of the following reactions does the value of  $\Delta H^\ominus$  represent **both** a standard enthalpy change of combustion and a standard enthalpy change of formation?

1.  $\text{C(s)} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$
2.  $2\text{C(s)} + \text{O}_2(\text{g}) \longrightarrow 2\text{CO(g)}$
3.  $\text{CO(g)} + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$

- ☐ None of the above
- ☐ 1 only
- ☐ 2 only
- ☐ 3 only
- ☐ 1 and 2 only
- ☐ 1 and 3 only
- ☐ 2 and 3 only
- ☐ All of the above

Adapted with permission from UCLES, A Level Chemistry, June 1994, Paper 4, Question 31

Question deck:

**STEM SMART Chemistry Week 13**

## Formation and Combustion Enthalpies 1

### Essential Pre-Uni Chemistry F3.1

**Subject & topics:** Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

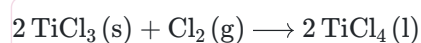
Data (all in  $\text{kJ mol}^{-1}$ ):

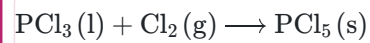
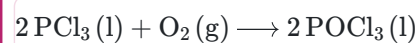
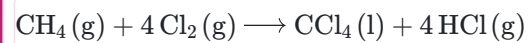
|                           | $\Delta_f H^\circ$ |                                                | $\Delta_c H^\circ$ |
|---------------------------|--------------------|------------------------------------------------|--------------------|
| $\text{CH}_4(\text{g})$   | −74.8              | $\text{C}_6\text{H}_6(\text{l})$               | −3267.4            |
| $\text{CCl}_4(\text{l})$  | −129.6             | $\text{H}_2(\text{g})$                         | −285.8             |
| $\text{HCl}(\text{g})$    | −92.3              | $\text{C}_6\text{H}_{12}(\text{l})$            | −3919.5            |
| $\text{TiCl}_4(\text{l})$ | −804.2             | $\text{C}_2\text{H}_2(\text{g})$               | −1300.8            |
| $\text{TiCl}_3(\text{s})$ | −720.9             | $\text{C}_2\text{H}_6(\text{g})$               | −1559.7            |
| $\text{PCl}_3(\text{l})$  | −319.7             | $\text{C}_2\text{H}_5\text{OH}(\text{l})$      | −1367.3            |
| $\text{PCl}_5(\text{s})$  | −443.5             | $\text{C}_2\text{H}_4(\text{g})$               | −1410.8            |
| $\text{POCl}_3(\text{l})$ | −597.1             | $\text{CH}_3\text{COOH}(\text{l})$             | −874.1             |
| $\text{GeO}(\text{s})$    | −212.1             | $\text{C}_6\text{H}_{14}(\text{l})$            | −4163.0            |
| $\text{GeO}_2(\text{s})$  | −551.0             | $\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})$ | −2237.9            |
| $\text{NH}_3(\text{g})$   | −46.1              | $\text{CO}(\text{g})$                          | −283.0             |
| $\text{TiO}_2(\text{s})$  | −939.7             | $\text{Mg}(\text{s})$                          | −601.7             |

Use standard enthalpies of formation to calculate the reaction enthalpies for the following reactions. Unless stated otherwise in the question part, give your answers to 1 decimal place.

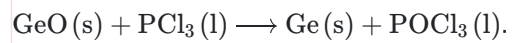
#### Part A

##### (a)

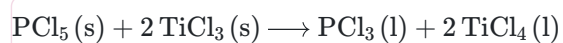


**Part B****(b)****Part C****(c)****Part D****(d)****Part E****(e)**

Part F  
(f)



Part G  
(g)



Part H  
(h)



Question deck:

**STEM SMART Chemistry Week 13**

## Formation and Combustion Enthalpies 2

### Essential Pre-Uni Chemistry F3.2

**Subject & topics:** Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

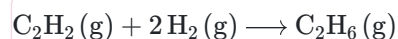
Data (all in  $\text{kJ mol}^{-1}$ ):

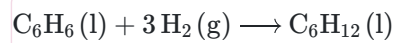
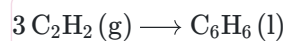
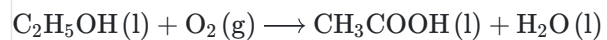
|                           | $\Delta_f H^\circ$ |                                                | $\Delta_c H^\circ$ |
|---------------------------|--------------------|------------------------------------------------|--------------------|
| $\text{CH}_4(\text{g})$   | −74.8              | $\text{C}_6\text{H}_6(\text{l})$               | −3267.4            |
| $\text{CCl}_4(\text{l})$  | −129.6             | $\text{H}_2(\text{g})$                         | −285.8             |
| $\text{HCl}(\text{g})$    | −92.3              | $\text{C}_6\text{H}_{12}(\text{l})$            | −3919.5            |
| $\text{TiCl}_4(\text{l})$ | −804.2             | $\text{C}_2\text{H}_2(\text{g})$               | −1300.8            |
| $\text{TiCl}_3(\text{s})$ | −720.9             | $\text{C}_2\text{H}_6(\text{g})$               | −1559.7            |
| $\text{PCl}_3(\text{l})$  | −319.7             | $\text{C}_2\text{H}_5\text{OH}(\text{l})$      | −1367.3            |
| $\text{PCl}_5(\text{s})$  | −443.5             | $\text{C}_2\text{H}_4(\text{g})$               | −1410.8            |
| $\text{POCl}_3(\text{l})$ | −597.1             | $\text{CH}_3\text{COOH}(\text{l})$             | −874.1             |
| $\text{GeO}(\text{s})$    | −212.1             | $\text{C}_6\text{H}_{14}(\text{l})$            | −4163.0            |
| $\text{GeO}_2(\text{s})$  | −551.0             | $\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})$ | −2237.9            |
| $\text{NH}_3(\text{g})$   | −46.1              | $\text{CO}(\text{g})$                          | −283.0             |
| $\text{TiO}_2(\text{s})$  | −939.7             | $\text{Mg}(\text{s})$                          | −601.7             |

Use standard enthalpies of combustion to calculate the reaction enthalpies for the following reactions:

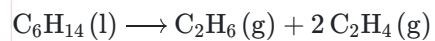
#### Part A

##### (a)

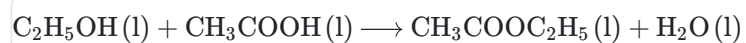


**Part B****(b)****Part C****(c)****Part D****(d)****Part E****(e)**

Part F  
(f)

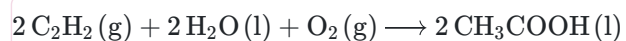


Part G  
(g)



Give your answer to 2 significant figures.

Part H  
(h)



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## Formation and Combustion Enthalpies 3

### Essential Pre-Uni Chemistry F3.3

**Subject & topics:** Chemistry | Physical | Energetics      **Stage & difficulty:** A Level P2

Data (all in  $\text{kJ mol}^{-1}$ ):

|                           | $\Delta_f H^\circ$ |                                                | $\Delta_c H^\circ$ |
|---------------------------|--------------------|------------------------------------------------|--------------------|
| $\text{CH}_4(\text{g})$   | −74.8              | $\text{C}_6\text{H}_6(\text{l})$               | −3267.4            |
| $\text{CCl}_4(\text{l})$  | −129.6             | $\text{H}_2(\text{g})$                         | −285.8             |
| $\text{HCl}(\text{g})$    | −92.3              | $\text{C}_6\text{H}_{12}(\text{l})$            | −3919.5            |
| $\text{TiCl}_4(\text{l})$ | −804.2             | $\text{C}_2\text{H}_2(\text{g})$               | −1300.8            |
| $\text{TiCl}_3(\text{s})$ | −720.9             | $\text{C}_2\text{H}_6(\text{g})$               | −1559.7            |
| $\text{PCl}_3(\text{l})$  | −319.7             | $\text{C}_2\text{H}_5\text{OH}(\text{l})$      | −1367.3            |
| $\text{PCl}_5(\text{s})$  | −443.5             | $\text{C}_2\text{H}_4(\text{g})$               | −1410.8            |
| $\text{POCl}_3(\text{l})$ | −597.1             | $\text{CH}_3\text{COOH}(\text{l})$             | −874.1             |
| $\text{GeO}(\text{s})$    | −212.1             | $\text{C}_6\text{H}_{14}(\text{l})$            | −4163.0            |
| $\text{GeO}_2(\text{s})$  | −551.0             | $\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})$ | −2237.9            |
| $\text{NH}_3(\text{g})$   | −46.1              | $\text{CO}(\text{g})$                          | −283.0             |
| $\text{TiO}_2(\text{s})$  | −939.7             | $\text{Mg}(\text{s})$                          | −601.7             |

Use enthalpies of formation and combustion to calculate the reaction enthalpy for the reaction:

$\text{Ge}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \longrightarrow \text{GeO}_2(\text{s}) + 2\text{H}_2(\text{g})$  Give your answer to 3 significant figures.

Question deck:

**STEM SMART Chemistry Week 13**



## Formation and Combustion Enthalpies 4

### Essential Pre-Uni Chemistry F3.4

**Subject & topics:** Chemistry | Physical | Energetics

**Stage & difficulty:** A Level P2

Data (all in  $\text{kJ mol}^{-1}$ ):

|                           | $\Delta_f H^\circ$ |                                                | $\Delta_c H^\circ$ |
|---------------------------|--------------------|------------------------------------------------|--------------------|
| $\text{CH}_4(\text{g})$   | −74.8              | $\text{C}_6\text{H}_6(\text{l})$               | −3267.4            |
| $\text{CCl}_4(\text{l})$  | −129.6             | $\text{H}_2(\text{g})$                         | −285.8             |
| $\text{HCl}(\text{g})$    | −92.3              | $\text{C}_6\text{H}_{12}(\text{l})$            | −3919.5            |
| $\text{TiCl}_4(\text{l})$ | −804.2             | $\text{C}_2\text{H}_2(\text{g})$               | −1300.8            |
| $\text{TiCl}_3(\text{s})$ | −720.9             | $\text{C}_2\text{H}_6(\text{g})$               | −1559.7            |
| $\text{PCl}_3(\text{l})$  | −319.7             | $\text{C}_2\text{H}_5\text{OH}(\text{l})$      | −1367.3            |
| $\text{PCl}_5(\text{s})$  | −443.5             | $\text{C}_2\text{H}_4(\text{g})$               | −1410.8            |
| $\text{POCl}_3(\text{l})$ | −597.1             | $\text{CH}_3\text{COOH}(\text{l})$             | −874.1             |
| $\text{GeO}(\text{s})$    | −212.1             | $\text{C}_6\text{H}_{14}(\text{l})$            | −4163.0            |
| $\text{GeO}_2(\text{s})$  | −551.0             | $\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})$ | −2237.9            |
| $\text{NH}_3(\text{g})$   | −46.1              | $\text{CO}(\text{g})$                          | −283.0             |
| $\text{TiO}_2(\text{s})$  | −939.7             | $\text{Mg}(\text{s})$                          | −601.7             |

Use the reaction enthalpies given, and the combustion or formation enthalpies above to find the requested enthalpy change in each case:

#### Part A



$\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \longrightarrow \text{NH}_4\text{Cl}(\text{s}), \Delta_r H^\circ = -176 \text{ kJ mol}^{-1}$  find  $\Delta_f H^\circ$  of  $\text{NH}_4\text{Cl}(\text{s})$

**Part B**

$\text{TiCl}_4 (\text{l}) + 2 \text{Mg} (\text{s}) \longrightarrow 2 \text{MgCl}_2 (\text{s}) + \text{Ti} (\text{s})$   $\Delta_r H^\circ = -478.4 \text{ kJ mol}^{-1}$ , find  $\Delta_f H^\circ$  of  $\text{MgCl}_2 (\text{s})$

**Part C**

$\text{CH}_3\text{COOCOCH}_3 (\text{l}) + \text{H}_2\text{O} (\text{l}) \longrightarrow 2 \text{CH}_3\text{COOH} (\text{l})$   $\Delta_r H^\circ = -46 \text{ kJ mol}^{-1}$ , find  $\Delta_c H^\circ$  of  $\text{CH}_3\text{COOCOCH}_3 (\text{l})$   
Give your answer to 4 significant figures.

**Part D**

$4 \text{C}_2\text{H}_2 (\text{g}) \longrightarrow \text{C}_6\text{H}_5\text{CHCH}_2 (\text{l})$ ,  $\Delta_r H^\circ = -808.2 \text{ kJ mol}^{-1}$ , find  $\Delta_c H^\circ$  of  $\text{C}_6\text{H}_5\text{CHCH}_2$  Give your answer to 4 significant figures.

**Part E**

$4 \text{Al} (\text{s}) + 3 \text{GeO}_2 (\text{s}) \longrightarrow 2 \text{Al}_2\text{O}_3 (\text{s}) + 3 \text{Ge} (\text{s})$   $\Delta_r H^\circ = -1698.4 \text{ kJ mol}^{-1}$ , find  $\Delta_f H^\circ$  of  $\text{Al}_2\text{O}_3 (\text{s})$  Give your answer to 4 significant figures.

**Part F**

$\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \longrightarrow 2\text{Fe}(\text{s}) + 3\text{CO}_2(\text{g})$ ,  $\Delta_r H^\circ = -24.8 \text{ kJ mol}^{-1}$ , find  $\Delta_f H^\circ$  of  $\text{Fe}_2\text{O}_3$

**Part G**

$3\text{CuO}(\text{s}) + 2\text{NH}_3(\text{g}) \longrightarrow 3\text{Cu}(\text{s}) + \text{N}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$ ,  $\Delta_r H^\circ = -293.3 \text{ kJ mol}^{-1}$ , find  $\Delta_f H^\circ$  of  $\text{CuO}(\text{s})$  Give your answer to 3 significant figures.

**Part H**

$2\text{PCl}_5(\text{s}) + 8\text{H}_2\text{O}(\text{l}) \longrightarrow 2\text{H}_3\text{PO}_4(\text{s}) + 10\text{HCl}(\text{g})$ ,  $\Delta_r H^\circ = -307.6 \text{ kJ mol}^{-1}$ , find  $\Delta_f H^\circ$  of  $\text{H}_3\text{PO}_4(\text{s})$  Give your answer to 3 significant figures.

**Part I**

$\text{Ga}_2\text{O}_3(\text{s}) + 3\text{Mg}(\text{s}) \longrightarrow 2\text{Ga}(\text{s}) + 3\text{MgO}(\text{s})$ ,  $\Delta_r H^\circ = -716.1 \text{ kJ mol}^{-1}$ , find  $\Delta_c H^\circ$  of Ga.

Part J

HCl(g)

$\text{TiCl}_4(\text{l}) + 2\text{H}_2\text{O}(\text{l}) \longrightarrow \text{TiO}_2(\text{s}) + 4\text{HCl}(\text{aq})$ ,  $\Delta_r H^\circ = -232.3 \text{ kJ mol}^{-1}$ , find  $\Delta_{\text{sol}} H^\circ$  of HCl(g) Give your answer to 3 significant figures.

Question deck:

**STEM SMART Chemistry Week 13**

## Reducing Carbon Dioxide

**Subject & topics:** Chemistry | Physical | Energetics      **Stage & difficulty:** A Level P1

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The standard enthalpy changes of formation of carbon monoxide and carbon dioxide are  $-110 \text{ kJ mol}^{-1}$  and  $-393 \text{ kJ mol}^{-1}$ , respectively.

### Part A

#### Carbon monoxide formation

Write an equation, including state symbols, for the first of these enthalpy changes (formation of carbon monoxide).

### Part B

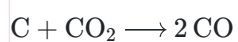
#### Carbon dioxide formation

Write an equation, including state symbols, for the second of these enthalpy changes (formation of carbon dioxide).

### Part C

#### Standard enthalpy change

Use the two standard enthalpy of formation values to calculate, in  $\text{kJ mol}^{-1}$ , the standard enthalpy change of the reaction



**Part D**  
**Condition**

In light of the result obtained in the previous part, suggest what condition is necessary to obtain a reasonable yield of carbon monoxide by this reaction.

- ☐ Low temperature
- ☐ Presence of  $O_2$
- ☐ Low pressure
- ☐ High temperature

Adapted with permission from UCLES, A Level Chemistry, June 1990, Paper 2, Question 1

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Question deck:

**STEM SMART Chemistry Week 13**

## C<sub>3</sub>H<sub>6</sub> Combustion

**Subject & topics:** Chemistry | Physical | Energetics      **Stage & difficulty:** A Level P2

**A** and **B** are two isomers with the molecular formula C<sub>3</sub>H<sub>6</sub>. The standard enthalpies of formation,  $\Delta_f H^\ominus$ , of both **A** and **B** have been found by first measuring the standard enthalpies of combustion,  $\Delta_c H^\ominus$ , of each. These values are given in the table below, together with the standard enthalpies of combustion of carbon and hydrogen.

|                                           | <b>A</b> | <b>B</b> | <b>carbon</b> | <b>hydrogen</b> |
|-------------------------------------------|----------|----------|---------------|-----------------|
| $\Delta_c H^\ominus / \text{kJ mol}^{-1}$ | −2058    | −2091    | −393.5        | −241.8          |

### Part A

#### Combustion equation

Give the equation for the complete combustion of C<sub>3</sub>H<sub>6</sub>. (Balance it for one mole of the hydrocarbon.)

### Part B

#### $\Delta_f H^\ominus$ of A

Calculate the standard enthalpy of formation of **A**.

**Part C**

$\Delta_f H^\ominus$  of **B**

Calculate the standard enthalpy of formation of **B**.

**Part D**

**Isomerisation**

Gaseous **B** needs to be stored carefully since it can convert explosively to the elements, to isomer **A**, or to other hydrocarbons. Calculate the standard enthalpy change for the reaction **B**  $\longrightarrow$  **A**.

Adapted with permission from the Cambridge Chemistry Challenge 2011, Question 1

Question deck:

**STEM SMART Chemistry Week 13**



## Ethene Combustion

**Subject & topics:** Chemistry | Physical | Energetics      **Stage & difficulty:** A Level C1

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The standard enthalpy change of combustion of but-1-ene,  $\text{C}_4\text{H}_8(\text{g})$ , is  $x \text{ kJ mol}^{-1}$ .

The standard enthalpy change of the reaction  $2 \text{C}_2\text{H}_4(\text{g}) \longrightarrow \text{C}_4\text{H}_8(\text{g})$  is  $y \text{ kJ mol}^{-1}$ .

Write down an expression, in terms of  $x$  and  $y$ , for the standard enthalpy change of combustion of ethene,  $\text{C}_2\text{H}_4(\text{g})$  when expressed in  $\text{kJ mol}^{-1}$  (your answer should not feature any units).

The following symbols may be useful:  $x$ ,  $y$

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Adapted with permission from UCLES, A Level Chemistry, November 1993, Paper 4, Question 8

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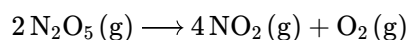
Question deck:

**STEM SMART Chemistry Week 13**

## Dinitrogen Pentoxide Formation

**Subject & topics:** Chemistry | Physical | Energetics      **Stage & difficulty:** A Level C2

In the gas phase,  $\text{N}_2\text{O}_5$  decomposes to oxygen and nitrogen dioxide. Use the data below (determined at 298 K) to calculate the standard enthalpy change at 298 K for the reaction:



|                                                                                                                      | value/ $\text{kJ mol}^{-1}$ |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------|
| $\Delta_f H^\circ$ of $\text{N}_2\text{O}_5(\text{g})$                                                               | 11.3                        |
| $\Delta_r H^\circ$ for $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{NO}_2(\text{g})$ | -58.1                       |
| bond strength in $\text{N}_2(\text{g})$                                                                              | 945                         |
| bond strength in $\text{O}_2(\text{g})$                                                                              | 498                         |
| bond strength in $\text{NO}(\text{g})$                                                                               | 631                         |

Adapted with permission from the Cambridge Chemistry Challenge, Question 1, 2019

Question deck:

**STEM SMART Chemistry Week 13**